NIRMAYEE S. KADAM PRN-25070521134 SECTION-C1

EXPERIMENT-8.1.1 MULTIPLICATION OF TWO SQUARE MATRICES

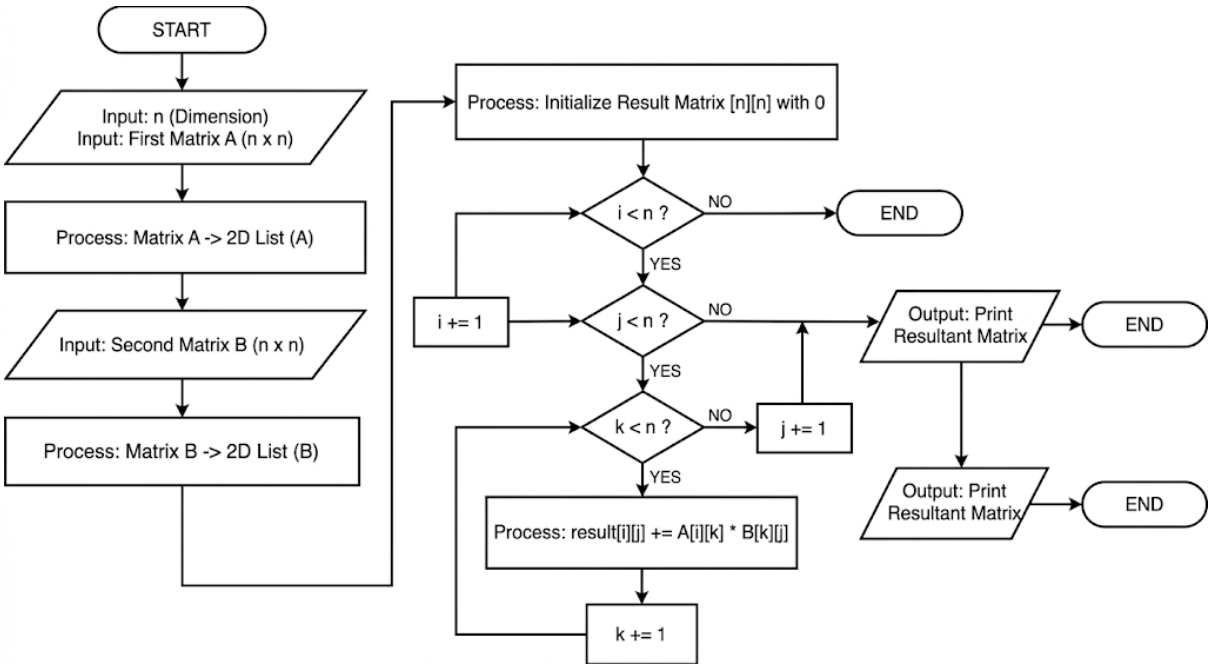
ALGORITHM:

1. Take size n.
2. Input two matrices
3. For each position:

- Multiply row of first matrix with column of second matrix
- Add the results

4. Store the value
5. Print the final matrix

FLOWCHART:



CODE:

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8.1.1. Multiplication of Two Square Matrices 02:21

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices $(n \times n)$.
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases +

Explorer

Multiplica... Submit

Debugger

```
1
2  n = int(input("dimension: "))
3
4
5  print("first matrix:")
6  A = []
7  for i in range(n):
8      row = list(map(int, input().split()))
```

Average time 0.019 s 19.25 ms

Maximum time 0.020 s 20.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 20 ms Debug

Expected output	Actual output
dimension: 2	dimension: 2
first matrix:	first matrix:
1 2	1 2

Terminal Test cases